

Bhanu Nagpure

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Android & Flutter Developer — On-Device AI · Production Apps · Open to On-site / Remote / Hybrid

Profile

Mobile engineer and indie app developer with **2+ years** of hands-on experience building and shipping production-grade Android and Flutter applications on Google Play. Founder of **Devanshu Studios** — independently conceived, built, and launched 2 apps, including an **on-device AI chatbot** powered by LiteRT-LM and a cross-platform Python learning platform with 1,200+ exercises. Deep expertise across **Kotlin, Jetpack Compose, Flutter, Dart, Riverpod 2.0, Clean Architecture, and on-device AI inference**. International research intern (Mitacs Globalink, Canada). Published researcher, 2-time patent applicant (1 granted), open-source contributor at Wikimedia Foundation, and technical writer on Medium.

Education

G H Raisoni College of Engineering

Bachelor of Technology in Computer Science & Engineering | CGPA: 8.7 / 10.0

Nagpur, India

Dec 2021 – Jun 2025

Technical Skills

Android: Kotlin, Java, Jetpack Compose (Material 3), Android SDK, MVVM/MVI, Hilt (DI), WorkManager, CameraX, Room, Proto DataStore, EncryptedSharedPreferences

Flutter & Dart: Flutter SDK, Dart, Riverpod 2.0, BLoC/Cubit, Clean Architecture, Flutter Flavors, go_router, Platform Channels

On-Device AI: LiteRT-LM (TFLite runtime), Gemma 2/3, Phi 3.5 Mini, SmolLM, MediaPipe FaceMesh, Google ML Kit

AI & ML: Gemini LLM (API & on-device), TensorFlow, PyTorch, Scikit-learn, BERT/Transformers, YOLOv5, OpenCV

Backend & Data: Firebase (Auth, Firestore, Storage, Crashlytics), REST APIs, OpenStreetMaps, Hive, SQL

Monetisation: RevenueCat (IAP/subscriptions), Google AdMob (native ad mediation), Play Console publishing

Languages: Kotlin, Dart, Python, Java, C++, SQL, JavaScript

Tools: Git, GitHub, Android Studio, Xcode, Gradle (Kotlin DSL), CI/CD, Google Cloud Platform, Linux

Professional Experience

Founder & Lead Mobile Engineer

Aug 2025 – Present

Devanshu Studios

Remote

- Conceived, built, and shipped **Chakuli — Offline AI Chatbot** (Android, closed beta on Play Store): a **privacy-first on-device AI assistant** running Gemma 2/3, Phi 3.5 Mini, and SmolLM entirely via **LiteRT-LM** — zero cloud calls for AI inference.
- Engineered the full **Jetpack Compose** (Material 3) UI, **Hilt DI**, **Proto DataStore**, **WorkManager** background model downloads, custom **CameraX** pipeline, Agent Skills system, and encrypted key storage.
- Independently architected and shipped **PyMaster** (Live on Play Store) — a **50k+ LOC Flutter** Python learning platform with Clean Architecture, **Riverpod 2.0**, Flutter Flavors, and a **Gemini AI Tutor**.
- Built an **offline-first sync engine**: Riverpod + Hive queues user progress locally and batch-uploads to Firebase on reconnect — zero data loss.
- Implemented full **monetisation stack**: RevenueCat (subscriptions & IAP), Google AdMob with native ad mediation, Firebase Google Sign-In — managed entire lifecycle from code to Play Console.
- Technical writer on Medium covering engineering decisions and production postmortems: “*I Didn’t Stress Test My Native Ads. My App Paid the Price.*” and “*Why I Stopped Building Apps and Started Building an Engine in Flutter.*”

Software Developer Intern

Oct 2022 – May 2024

BETIC-GHRCE (Business & Entrepreneurship Technology Innovation Centre)

Nagpur, India

- Built a production **Android (Kotlin/Java)** app for TMJ disorder screening — integrated **MediaPipe FaceMesh API** for real-time facial landmark detection, improving diagnostic classification accuracy by **60%**.

- Optimised **Google ML Kit** camera pipeline, achieving a **30%** performance boost and enabling sub-second inference for clinical use.
- Contributed to patent filing: “*Smart Chairside TMJ Examination and Muscle Activity Detector Tool*” (Pub. No. 202321022476, Filed Mar 2023 — **Granted**).

Open Source Contributor

Feb 2025 – Present

Wikimedia Foundation — Scribe-Android

Remote

- Developed tablet-specific keyboard layouts in **Kotlin** for Scribe-Android, covering 7+ languages. Resolved critical UI bugs and refactored the core **KeyHandler** module for improved maintainability.

Machine Learning Intern

Aug 2024 – Oct 2024

Athabasca University

Athabasca, AB, Canada

- Delivered **95% prediction accuracy** for a text ideology identification model using **Transformers / BERT**; deployed a prototype web interface for researcher validation and experiment reproduction.

Globalink Research Intern

Apr 2024 – Aug 2024

Mitacs

Edmonton, AB, Canada

- Led end-to-end development of a **cross-platform (iOS, Android, Web)** field-trip application as sole developer — Flutter frontend, Firebase backend, and OpenStreetMaps integration.
- Authored full technical documentation; presented research at the **3rd Annual TRESL Lab International Symposium**, Athabasca, Canada (Jul 2024).

Fellowship — Design for Health

Mar 2022 – Oct 2022

IIT Bombay

Mumbai, India

- Led a multidisciplinary team to prototype a medical device for xerostomia, reducing diagnostic time by **80%**. Built using MIT App Inventor and IoT hardware; commended by medical and engineering experts at IIT Bombay.

Apps Shipped — Google Play

Chakuli — Offline AI Chatbot

Aug 2025 – Present

Kotlin, Jetpack Compose, LiteRT-LM, Hilt, CameraX, Proto DataStore

- **Closed beta on Google Play (12+ testers)**. Privacy-first Android AI assistant — Gemma 2/3, Phi 3.5 Mini, SmolLM running entirely on-device via LiteRT-LM. No internet required for AI inference.
- Agent Skills system, audio chat, in-app camera (CameraX), foreground WorkManager for model downloads, Protobuf DataStore for persistence, RevenueCat tip jar, opt-in Crashlytics.

PyMaster — Learn Python Coding

Aug 2025 – Present

Flutter, Dart, Riverpod 2.0, Firebase, RevenueCat, Gemini AI

- **Live on Google Play**. 1,200+ interactive exercises, gamified league system, Gemini-powered AI Tutor, offline-first sync — 50k+ LOC codebase built from scratch.
- Flutter Flavors (dev/staging/prod), Riverpod 2.0, RevenueCat (subscriptions & IAP), Google AdMob native mediation, Firebase Auth (Google Sign-In), Crashlytics.

Research & Academic Projects

OCanPredict — OSMF Screening App

Dec 2022 – Jul 2023

Android, Kotlin, Java, MediaPipe FaceMesh

- Android diagnostic app using **MediaPipe FaceMesh API** for real-time OSMF facial screening, increasing efficiency by **40%**. Won multiple national and state-level awards; contributed to granted patent.

V-Safe-Anywhere: Women’s Safety Wearable

Nov 2022 – Apr 2024

Flutter, Firebase, IoT, Google Cloud

- Flutter app paired with a custom IoT wearable: panic button triggers instant **live GPS broadcast** to emergency contacts via Firebase — **98% uptime**. Published by IGI Global.

Autonomous Garbage Collection Robot

Jun 2023 – Sep 2023

Python, YOLOv5, OpenCV, NumPy, IoT

- Computer vision system (YOLOv5) for autonomous floating waste detection — reduced manual effort by **60%**. Published at 8th ICACS 2024, ACM, Hong Kong.

Mobile Field Trip Application

Apr 2024 – Aug 2024

Flutter, Firebase, OpenStreetMaps

- Cross-platform (iOS, Android, Web) Flutter app for academic field assignments at Athabasca University — Firebase Auth/Storage backend, OpenStreetMaps integration. Presented at TRESL Lab Symposium, Canada.

Publications, Patents & Conferences

Journal: “V-Safe-Anywhere: Empowering Women’s Safety with Wearable AI and IoT.” *IGI Global*, Mar 2024. [\[Link\]](#)
Conference: “Autonomous Floating Garbage Collection Device using Computer Vision and Robotics.” *8th International Conference on Advances in Computing & Software Engineering (ICACS 2024)*, ACM, Hong Kong, Oct 2024. [\[Link\]](#)

Conference: “Development of a Mobile Field Trip Application.” *3rd Annual TRESL Lab International Symposium*, Athabasca, Canada, Jul 2024.

Patent (Granted): “*Smart Chairside TMJ Examination and Muscle Activity Detector Tool.*” Office of the Controller General of Patents, India. (Pub.No. 202321022476, Filed Mar 2023 — **Granted.**)

Patent (Pending): “*An Artificial Intelligence Based Portable Device for Dental Disease Detection.*” Govt. of India — DPIIT. (App.No. 202421006731, Filed Feb 2024, Published Aug 2025 — Awaiting Examination.)

Awards & Recognition

3rd Place — Inter-University Research Convention: Avishkar 2024, Nashik (Jan 2024)

Runner-up — DIPEX 2023, Amravati **Runner-up** — SHRISTI 2023, Kottayam

Grant Awardee — BIREC E-YUVA Grant, Rs. 10 lakh seed funding (Feb 2023)

Winner — MEDHA Innovation Challenge 2022 **Winner** — MEDIC-2022 (Oct 2022)

Multiple national and state-level awards for OCanPredict (OSMF screening) innovation.

Writing & Open Source

Medium Blog (medium.com/@bhanu.r.nagpure.8082): Technical articles on indie Android/Flutter development:

- “*I Built a Python Learning App From a Hospital Bed.*” (Jun 2026)
- “*I Didn’t Stress Test My Native Ads. My App Paid the Price.*” (Apr 2026)
- “*Why I Stopped Building Apps and Started Building an Engine in Flutter.*” (Mar 2026)

Open Source: Contributor to **Wikimedia Foundation** / **Scribe-Android** (Kotlin) — tablet keyboard layouts for 7+ languages, UI bug fixes, KeyHandler refactor.